

SSLC KARNATAKA**Mathematics****March- 2015****Class-X Time: 3 Hours****Max. Marks: 80**

General Instructions to the Candidate:

- (i) This Question Paper consists of 40 objective and subjective of questions.
- (ii) This questions paper has been sealed by reverse jacket. You have to cut on the right side to open the paper at the time of commencement of the examination .Check whether all the pages of the question paper are intact.
- (iii) Follow the instructions given against both the objective and subjective types of questions.
- (iv) Figures in the right hand margin indicate maximum marks for the questions.
- (v) The maximum time to answer the paper is given at the top the question paper. It includes 15 for minutes for reading the question paper.

SECTION - A

I. Four alternatives are given for each of the following questions / incomplete statements. Only one of them is correct or most appropriate. Choose the correct alternative and write the complete answer along with its letter in the space provided against each question.

1. If $A = \{ a, b, c \}$ and $B = \{ c, d, e \}$, then $A \cap B$ is

- (A) $\{ a, b, c, d, e \}$ (B) $\{ c \}$
(C) $\{ a, b \}$ (D) $\{ d, e \}$

Ans. $\{ c \}$

2. In a geometric progression, if $T_4 = 10$ and common ratio is 2 then T_3 is

- (A) 5 (B) 20
(C) 8 (D) 12.

Ans. 5

3. Which of the following is an identity (unit) matrix?

- (A) $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$ (B) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(C) $\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Ans. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

4. The value of ${}^5P_1 - {}^3P_0$ is

(A) 4 (B) 2

(C) 3 (D) 0

Ans. 4

5. The HCF and LCM of two expressions A and B are H and L respectively. Then their relation can be written as

(A) $A \times H = B \times L$ (B) $A \times B = H \times L$

(C) $\frac{A}{H} = \frac{B}{L}$ (D) $\frac{A}{L} = \frac{B}{H}$

Ans. $A \times B = H \times L$

6. The L.C.M of two expressions $a^2 - 2ab + b^2$ and $a^2 - b^2$ is $(a - b)^2(a + b)$. Their H.C.F is

(A) $(a + b)$ (B) $(a - b)$

(C) $(a - b)^2$ (D) $(a^2 - b^2)$

Ans. $(a - b)$

7. The value of $\sum_{a,b,c} (a - b)$ is

(A) $a + b + c$ (B) 0

(C) $a - b - c$ (D) $a - b$

Ans. (b)

8. The factors of $a^3 + b^3$ are

(A) $(a + b)(a^2 + ab + b^2)$ (B) $(a + b)(a^2 + ab - b^2)$

(C) $(a + b)(a^2 - ab + b^2)$ (D) $(a + b)(a^2 - ab - b^2)$

Ans. $(a + b)(a^2 - ab + b^2)$

9. The product of $\sqrt[3]{2}$ and $\sqrt[3]{3}$ is

- (A) $\sqrt[6]{6}$ (B) $\sqrt[6]{9}$ (C) $\sqrt[6]{72}$ (D) $\sqrt[6]{27}$

Ans. $\sqrt[6]{9}$

10. Which of the following is a pure quadratic equation?

- (A) $x(x - 2) = 0$ (B) $x + \frac{1}{x} = 5$
(C) $x + \frac{1}{x} = 0$ (D) $(x - 2)^2 = 0$

Ans. $(x - 2)^2 = 0$

11. Square of a number is added to three times the same number and their sum is 28. This statement can be written in the equation form as

- (A) $3a^2 + 3a + 28 = 0$ (B) $a^2 = 3a + 28$
(C) $a^2 + 3a = 28$ (D) $3a^2 + a = 28$

Ans. $a^2 + 3a = 28$

12. The product of the roots of the equation $x^2 - 5x = 0$ is

- (A) 0 (B) 5
(C) -5 (D) 1

Ans. 0

13. The nature of the roots of the equation $x^2 - 4x + 4 = 0$ is

- (A) Real and equal (B) real and rational
(C) Real and irrational (D) imaginary.

Ans. Real and equal

14. Circles having same centre but different radii are called

- (A) Concentric circles (B) congruent circles
(C) Intersecting circles (D) touching circles.

Ans. Concentric circles

15. A straight line drawn parallel to the side of a triangle divides the other two sides proportionally. This statement was enunciated by

- (A) Thales (B) Pythagoras
(C) Baudhayana (D) Appolonius.

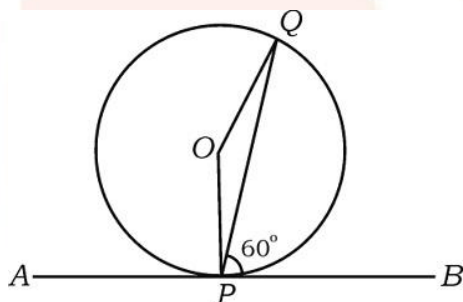
Ans. Pythagoras

16. Angle between radii at the centre of a circle is 120° . Then the angle between tangents drawn at the ends of the radii is

- (A) 60° (B) 120°
(C) 90° (D) 100° .

Ans. 60°

17. In the figure APB is a tangent at P to the circle with centre O. If $\angle QPB = 60^\circ$ then $\angle POQ$ is



- (A) 60° (B) 100°
(C) 120° (D) 150°

Ans. 120°

18. The circumference of the base of a right circular cylinder is 44 cm and height is 10 cm. Then its lateral surface area is

- (A) 44 cm^2 (B) 400 cm^2 (C) 54 cm^2 (D) 440 cm^2

Ans. 440 cm^2

19. The formula used to find the total surface area of a cone is

- (A) $\pi r l$ (B) $\pi r (r + l)$
(C) $2\pi r h$ (D) $2\pi r (r + h)$

Ans. $\pi r (r + l)$

20. The shape of each face of a cube is

- (A) square (B) triangular
 (C) pentagonal (D) hexagonal.

Ans. Square

II. Fill in the blanks with suitable answers:

21. If 2, 4 and 8 are in geometric progression then the common ratio is..... .

Ans. 2

22. If A, G, H are Arithmetic mean, Geometric mean and Harmonic mean respectively then $\sqrt{AH}$ is..... .

Ans. Geometric mean

24. In a right angled triangle, the side opposite to right angle is called.....

Ans. Hypotenuse

25. A solid obtained by the rotation of a rectangle about one of its fixed sides is called.....

Ans. Cylinder

26. If $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ then $A - A'$ is.....

Ans. $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

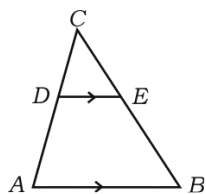
27. If $\sum_{a,b,c} a = 0$ then the value of $\sum_{a,b,c} (a)^3$ is.....

Ans. 0

28. If the HCF of ab^2 and a^2b is ab then the product of HCF and LCM is..... .

Ans. a^3b^3

29. In the figure $DE \parallel AB$. If $AD = 7$, $CD = 5$ and $BC = 18$ then CE is.....



Ans. 7.5cm

30. Two circles of radii 5 cm and 3 cm touch each other internally. Then the distance between their centres is.....

Ans. 8cm

III. 31. If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5\}$, then find $A - B$ and draw Venn diagram.

Ans. Given, $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5\}$

$$\therefore A - B = \{1, 2\}$$

32. A florist has certain number of garlands. 120 of them have Champak, 60 have Jasmine flowers and 30 garlands have both the flowers. Find the total number of garlands with him.

Ans. Let A be the set having garlands having Champak, therefore $n(A) = 120$.

Let B be the set of garlands having Jasmine, therefore $n(B) = 60$.

$A \cap B$ is the set of garlands which has both these flowers.

Therefore $(A \cap B) = 30$.

$A \cup B$ is the set of garlands the florist has.

We know,

$$\begin{aligned} n(A \cup B) &= n(A) + n(B) - n(A \cap B) \\ &= 120 + 60 - 30 \\ &= 150 \end{aligned}$$

Therefore, the florist has 150 garlands.

33. Three terms are in geometric progression. Their product is 8 and common ratio is 2, then find the first term.

Ans. Let the number in geometric progression be $\frac{a}{r}$, a and ar .

Given, common ratio (r) = 2

According to question,

Product of numbers = 8

$$\Rightarrow \frac{a}{r} \times a \times ar = 8$$

$$\Rightarrow a^3 = 8$$

$$\therefore a = 2$$

Hence, the first term is $\frac{a}{r} = \frac{2}{2} = 1$

34. Find T_{10} in the harmonic progression $\frac{1}{2}, \frac{1}{5}, \frac{1}{8}, \dots$

Ans. Given sequence is $\frac{1}{2}, \frac{1}{5}, \frac{1}{8}, \dots$

Here, first term = $\frac{1}{a} = \frac{1}{2}$,
 $\therefore a = 2$

Let d be the common difference, then

Second term = $\frac{1}{5}$

$$\Rightarrow \frac{1}{a+d} = \frac{1}{5}$$

$$\Rightarrow \frac{1}{2+d} = \frac{1}{5}$$

$$\Rightarrow 5 = 2 + d$$

$$\therefore d = 3$$

Now,

10th term (T_{10}) =

$$\frac{1}{[a + (n-1)d]}$$

$$= \frac{1}{[2 + (10-1)3]}$$

$$= \frac{1}{2+27}$$

$$= \frac{1}{29}$$

35. If $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 \\ -1 & 3 \end{pmatrix}$, then find AB.

Ans. Given, $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

$$B = \begin{pmatrix} 2 & 0 \\ -1 & 3 \end{pmatrix}$$

$$\therefore AB = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \times \begin{pmatrix} 2 & 0 \\ -1 & 3 \end{pmatrix}$$

$$= \begin{bmatrix} (1 \times 2 + 2 \times -1) & (1 \times 0 + 2 \times 3) \\ (3 \times 2 + 4 \times -1) & (3 \times 0 + 4 \times 3) \end{bmatrix}$$

$$= \begin{bmatrix} 2-2 & 0+6 \\ 6-4 & 0+12 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 6 \\ 2 & 12 \end{bmatrix}$$

36. Find the value of

(a) nP_0

(b) ${}^{10}P_1$

(c) ${}^{100}C_0$

(d) ${}^{2015}C_{2015}$

Ans. a) ${}^nP_0 = \frac{n!}{(n-0)!} = \frac{n!}{n!} = 1$

b) ${}^{10}P_1 = \frac{10!}{(10-1)!} = \frac{10!}{9!} = 10$

c) ${}^{100}C_0 = \frac{100!}{(100-0)!0!} = \frac{100!}{100!} = 1$

$$d) {}^{2015}C_{2015} = \frac{2015!}{(2015-2015)!2015!} = \frac{2015!}{2015!} = 1$$

37. Write $\sqrt[3]{5}$ in its exponential form and mention the rational factor, order and radicand.

Ans. The exponent form of $\sqrt[3]{5}$ is $5^{\frac{1}{3}}$

The rational factor of $\sqrt[3]{5}$ is 125. Its order is third and radicand is 5.

38. Rationalise the denominator and simplify: $\frac{\sqrt{2}}{\sqrt{5}-\sqrt{3}}$

$$\begin{aligned} \text{Ans. } & \frac{\sqrt{2}}{\sqrt{5}-\sqrt{3}} \\ &= \frac{\sqrt{2}}{\sqrt{5}-\sqrt{3}} \times \frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}+\sqrt{3}} \\ &= \frac{\sqrt{2}(\sqrt{5}+\sqrt{3})}{(\sqrt{5}-\sqrt{3})(\sqrt{5}+\sqrt{3})} \\ &= \frac{\sqrt{10}+\sqrt{6}}{(\sqrt{5})^2-(\sqrt{3})^2} \\ &= \frac{\sqrt{10}+\sqrt{6}}{5-3} \\ &= \frac{\sqrt{10}+\sqrt{6}}{2} \end{aligned}$$

39. Express $2x^2 = 5x - 3$ in the standard form of quadratic equation and write the values of a , b and c .

Ans. Given equation is $2x^2 = 5x - 3$

The standard form of given quadratic equation is $2x^2 - 5x + 3 = 0$

Comparing it with $ax^2 + bx + c = 0$ we get

$a = 2$, $b = -5$ and $c = 3$.

40. The product of two consecutive integers is 12. Find the numbers.

Ans. Let the numbers be x and $x+1$.

According to question,

$$x(x+1) = 12$$

$$\Rightarrow x^2 + x - 12 = 0$$

$$\Rightarrow x^2 + 4x - 3x - 12 = 0$$

$$\Rightarrow x(x+4) - 3(x+4) = 0$$

$$\Rightarrow (x+4)(x-3) = 0$$

Either,

$$x+4 = 0$$

$$\therefore x = -4$$

Or

$$x-3 = 0$$

$$\therefore x = 3$$

Hence, the required numbers are 3 and 4.

41. Solve $x^2 + 2x - 1 = 0$ by using formula.

Ans. Given equation is $x^2 + 2x - 1 = 0$

Comparing it with $ax^2 + bx + c = 0$, we get

$a = 1$, $b = 2$ and $c = -1$

We know,

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-2 \pm \sqrt{2^2 - 4 \times 1 \times -1}}{2 \times 1} \\ &= \frac{-2 \pm \sqrt{4 + 4}}{2} \\ &= \frac{-2 \pm \sqrt{8}}{2} \\ &= \frac{-2 \pm 2\sqrt{2}}{2} \\ &= \frac{-2 + 2\sqrt{2}}{2} \text{ or } \frac{-2 - 2\sqrt{2}}{2} \\ &= \frac{-2 + 2.83}{2} \text{ or } \frac{-2 - 2.83}{2} \\ &= 0.4 \text{ or } -2.4 \end{aligned}$$

42. Form a quadratic equation whose roots are 3 and 2.

Ans. Given, roots of a quadratic equation are 3 and 2.

Now,

$$\text{Sum of roots} = 3+2= 5$$

$$\text{Product of roots} = 2*3= 6$$

Hence,

$$\text{The Required equation will be } x^2 - 5x + 6 = 0$$

43. Construct Cayley's table $(Z_4, \oplus_4)$ under addition modulo 4.

44. Construct a tangent to a circle of radius 3 cm at any point P on it.

Ans. Step of construction:

1. Draw a circle of radius 3 cm with centre O.
2. Take a point P on the circle and Join OP.
3. Make a perpendicular lines at point P such that $AB \perp OP$.
4. Line AB would be tangent of circle at point P.

45. Write the formula for

- (a) Total surface area of cylinder
- (b) Volume of hemisphere.

Ans. a) Total Surface Area of cylinder = $2\pi rh$

Where, r = radius of the base of cylinder
 h = height of the cylinder

$$\text{b) Volume of hemisphere} = \frac{2}{3}\pi r^3$$

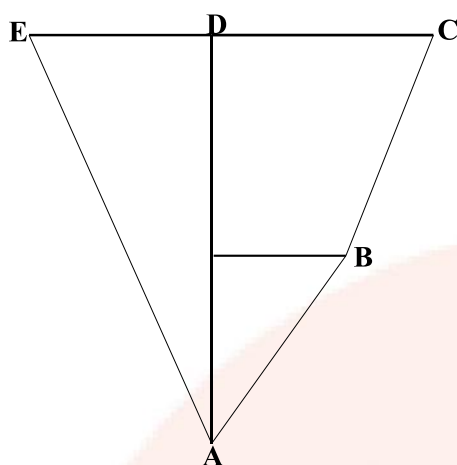
Where, r = radius of hemisphere

46. Draw a plan using the information given below:

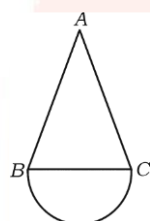
[Scale: 50 m = 1 cm]

	To D (in mts)	
To E 100	300 200 150 50	150 to C 100 to B
	From A	

Ans:



47. Verify Euler's formula for the graph:



Ans. In the given figure,
 Number of vertices (V) = 3
 Number of Faces (F) = 1
 Number of Edges (E) = 3

According to Euler's formula,
 $V + F - E = 2$

Substituting the values in the formula, we get

$$3 + 1 - 3 = 2$$

$$1 \neq 2$$

Hence, Euler's formula is not verified.

48. Draw a graph (network) for the matrix:

$$\begin{bmatrix} 2 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 0 \end{bmatrix}$$

IV. 49. There are 7 non-collinear points. How many (a) straight lines and (b) triangles can be drawn by joining these points?

Ans. a) We have 7 non collinear points.

So,

Number of lines that can be drawn

$$= {}^nC_2$$

$$= {}^{15}C_2$$

$$= 15 \times \frac{14}{2}$$

$$= 105$$

b) A triangle is formed by joining any three non-collinear points in pairs.

There are 7 non-collinear points.

The number of triangles formed,

$$= {}^7C_3$$

$$= \left[\frac{7 \times (7-1) \times (7-2)}{3!} \right]$$

$$= \frac{7 \times 6 \times 5}{3 \times 2 \times 1}$$

$$= 7 \times 5$$

$$= 35.$$

50. Calculate the standard deviation for the given frequency distribution table :

Class-interval	Frequency
1-5	2
6-10	3
11-15	4
16-20	1

Ans.

Class - interval	Frequency(f)	Mid-point(x)	fx	$(x - \bar{x})^2$	$f(x - \bar{x})^2$
1-5	2	3	6	$(3 - 10)^2 = 49$	98
6-10	3	8	24	$(8 - 10)^2 = 4$	12
11-15	4	13	52	$(13 - 10)^2 = 9$	36
16-20	1	18	18	$(18 - 10)^2 = 64$	64
	$\sum f = 10$		$\sum fx = 100$		$\sum f(x - \bar{x})^2 = 210$

Here, Mean ($\bar{x}$) = $\frac{\sum fx}{\sum f} = \frac{100}{10} = 10$

$$\therefore \text{Standard deviation} = \sqrt{\frac{\sum f(x - \bar{x})^2}{\sum f}} = \sqrt{\frac{210}{10}} = 21$$

51. Find the H.C.F of $4x^3 - 3x^2 - 24x - 9$ and $8x^3 - 2x^2 - 53x - 39$.

Ans. Given,

$$\begin{aligned} \text{First expression} &= 4x^3 - 3x^2 - 24x - 9 \\ &= 4x^2(x - 3) + 9x(x - 3) + 3(x - 3) \\ &= (x - 3)(4x^2 + 9x + 3) \end{aligned}$$

$$\text{Second expression} = 8x^3 - 2x^2 - 53x - 39.$$

$$\begin{aligned} &= 8x^2(x - 3) + 22x(x - 3) + 13(x - 3) \\ &= (x - 3)(8x^2 + 22x + 13) \end{aligned}$$

Hence, the H.C.F is $(x - 3)$.

52. Prove that —tangents drawn to a circle from an external point are equal.

Ans. Given:- PQ and TQ are two tangent drawn from an external point T to the circle C

To prove:- $PT = TQ$

$\angle OPT = \angle OTQ$

Construction: join OT

Proof:-

We know that, a tangent to circle is perpendicular to the radius through the point of contact.

So, $\angle OPT = \angle OQT = 90^\circ$

In $\triangle OPT$ and $\triangle OQT$,

$OT = OT$ (common side)

$OP = OQ$ (radius of the circle)

$\angle OPT = \angle OQT$ (90°)

So, $\triangle OPT \cong \triangle OQT$ (RHS)

So,

$PT = TQ$ and $\angle OPT = \angle OTQ$

Hence,

$PT = TQ$ [Since, the lengths of the tangent drawn from an external point to a circle are equal.]

$\angle OPT = \angle OTQ$ [Since, centre lies on the bisector of the angle between the two tangent.

Hence proved.

53. If $x^2 - 3x + 1 = 0$, then find the value of $x^2 + \frac{1}{x^2}$.

Ans. Given,

$$x^2 - 3x + 1 = 0$$

$$\Rightarrow x^2 + 1 = 3x \dots\dots(i)$$

Now,

$$x^2 + \frac{1}{x^2}$$

$$= \left(x + \frac{1}{x} \right)^2 - 2$$

$$= \left(\frac{x^2 + 1}{x} \right)^2 - 2$$

$$= \left(\frac{3x}{x} \right)^2 - 2 \quad [\text{From (i)}]$$

$$= 3^2 - 2$$

$$= 7$$

54. Show that the area of an equilateral triangle is $\frac{\sqrt{3}}{4} a^2$.

Ans. Let ABC be an equilateral triangle with sides 'a'. Now, draw AD perpendicular to BC. Let h be the height.

Here, we have $\triangle ABD = \triangle ADC$.

In $\triangle ABD$,

Using Pythagoras theorem,

$$a^2 = h^2 + \left(\frac{a}{2} \right)^2$$

$$\Rightarrow h^2 = a^2 - \frac{a^2}{4}$$

$$\Rightarrow h^2 = \frac{3a^2}{4}$$

$$\therefore h = \frac{\sqrt{3}}{2} a \dots\dots(i)$$

Now,

$$\text{Area of triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times a \times \frac{\sqrt{3}}{2} a$$

$$= \frac{\sqrt{3}}{4} a^2$$

Hence, the area of an equilateral triangle is $\frac{\sqrt{3}}{4} a^2$.

V.55. Draw direct common tangents to two circles of radii 4 cm and 2 cm with their centres 9 cm apart and measure their lengths.

Ans. Given, Radii of two circles are

$$R = 4\text{cm and } r = 2\text{cm}$$

$$\text{Distance between centre of the circles (d)} = 9\text{cm}$$

Let t be the length of the common tangent.

We know,

$$t = \sqrt{d^2 - (R - r)^2}$$

Substituting it the given values, we get

$$t = \sqrt{9^2 - (4 - 2)^2}$$

$$\Rightarrow t = \sqrt{81 - 2^2}$$

$$\Rightarrow t = \sqrt{78}$$

$$\therefore t = 8.8\text{cm}$$

Hence, the required length of tangent is 8.8cm

56. Prove that —if two triangles are equiangular then their corresponding sides are proportionall.

Ans. Construction: Two triangles ABC and DEF are drawn so that their corresponding angles are equal. This means:

$$\angle A = \angle D, \angle B = \angle E \text{ and } \angle C = \angle F$$

$$\text{To prove: } \frac{AB}{DE} = \frac{AC}{DF} = \frac{BC}{EF}$$

Draw a line PQ in the second triangle so that DP = AB and PQ = AC

Proof:

Since corresponding sides of these two triangles are equal,

$$\therefore \triangle ABC \cong \triangle DEF$$

So,

$$\angle B = \angle P = \angle E \text{ and } PQ \parallel EF$$

$$\therefore \frac{DP}{PE} = \frac{DQ}{QF}$$

$$\Rightarrow \frac{AB}{DE} = \frac{AC}{DF} \dots\dots(i)$$

Similarly,

$$\frac{AB}{DE} = \frac{BC}{EF} \dots\dots(ii)$$

Therefore,

$$\frac{AB}{DE} = \frac{AC}{DF} = \frac{BC}{EF} \quad [\text{From (i) and (ii)}]$$

Hence proved.

57. Find three numbers in Arithmetic Progression, whose sum and product are 6 and 6 respectively.

Ans. Let the three numbers be $a-d$, a , $a+d$

According to question,

Sum of numbers = 6

$$a-d + a + a+d = 6$$

$$a+a+a = 6$$

$$3a=6$$

$$\therefore a=2$$

Also,

Product of numbers = 6

$$(a-d) \times a \times (a+d) = 6$$

$$(a^2 - d^2) \times a = 6 \quad [\because (a+b)(a-b) = a^2 - b^2]$$

Substituting the value of a , we get

$$(2^2 - d^2) \times 2 = 6$$

$$4 - d^2 = 3$$

$$d^2 = 1$$

$$\therefore d = \sqrt{1} = +1 \text{ or } -1$$

Therefore,
If,
 $a=2$ and $d=1$

A.P will be 2,3,4,5,6.....
Or

If $a=2$ and $d=-1$
A.P will be 2,1,0,-1,-2.....

